ONE-SLIDE PROPOSAL · RIDES · DROP-OFF ACCURACY

Pathao drops commuters on the wrong road. Smart Drop-off fixes the last 100 metres.

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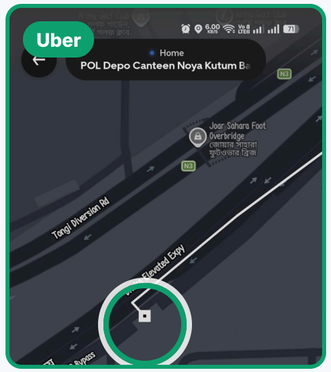
Same office, same 9:25 a.m. request: **Uber's route ends at the canteen**. **Pathao's ends across a divided highway**, on about half of my weekday trips.

1 THE MOMENT WE'RE LOSING

POL Depo Canteen Noya Kutum Bari, Dhaka



Pin on the canteen, route on the highway. It ends opposite the canteen.

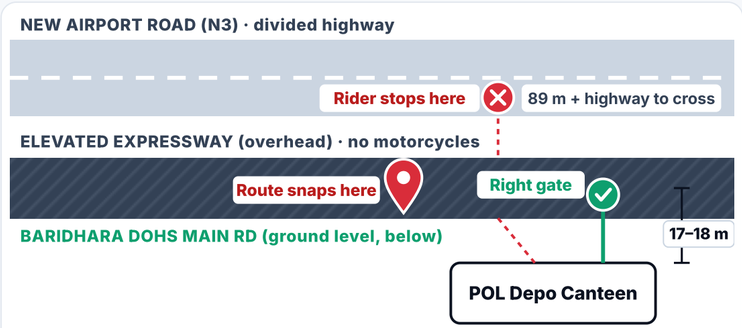


Route turns off the highway and ends at the canteen.

~1 in 2 of my office trips end on the highway. I cross a divided road on foot, or I switch to Uber.

2 WHY IT HAPPENS

The pin is right. The route ends on the wrong road.



- The expressway deck sits **almost exactly above** the canteen's road: 18 m vs 17 m away. On a flat map they are one line, so the route's end snaps to the **expressway**.
- Bikes can't use it, so navigation sends the rider to the **nearest legal road, N3**. The rider stops there, across the highway.
- The right road costs the rider only **+52 m and ~40 s**: one left turn off N3. Miss that turn and it's a 2.5 km detour.

Hypothesis built from OpenStreetMap road data and OSRM routes. Confirm with Pathao's geocoder and trip logs.

3 THE FIX: SMART DROP-OFF

Better GPS logic on the backend, plus coordination between passenger and rider

A — End the ride where bikes can stop BACKEND
Even when the pin is right, never end the route on a road bikes can't use: expressways, ramps, no-motorcycle roads. Use the nearest road they can.

B — Saved spot + your own tip PASSENGER
The passenger confirms the spot with one tap and can add a tip in their own words. The app asks to remember both, so next time there's no question.

C — Rider sees it before accepting RIDER
The spot and the tip come with the ride request, plus a short Bangla reminder. If the spot is on a highway, the rider is reminded to stop safely.

① Backend fix → ② Saved spot + tips → ③ Learn from trips

WHY MANAGEMENT SHOULD APPROVE IT

- **It protects commuters, our most frequent riders.** A daily route that fails half the time teaches them to open Uber first. At 2 trips a day, that's ~22 bad drop-offs a month for one person.
- **It's small and proven.** It uses map data and trip GPS we already have, step ① ships silently on the server, and food delivery apps already save rider notes per address.
- **Better GPS alone won't fix it.** The pin is already right. The route end and the way in are the problem, at every venue beside a flyover or expressway.

HOW WE'LL KNOW IT WORKED

- **Primary:** drop-off error, the distance from where the trip ended to the confirmed destination. Target: median under 25 m.
- **Also:** "wrong drop-off" tickets, rider–passenger calls in the last 2 minutes, and 30-day commuter retention.
- **Guardrails:** rider trip time, fare and acceptance rate. 4-week A/B test in Dhaka.

OPEN QUESTIONS BEFORE THIS SHIPS

- How many Dhaka destinations snap to no-stop roads? One trip-log query sizes it.
- Does our map provider give us bridge, layer and access tags, or must we derive them?
- Can riders enter every gate (cantonment and DOHS rules)? Who checks gates that passengers suggest?
- Does the rider need an incentive for the extra turn-in?

Screenshots: my own trips, same destination, same time (9:25). Distances: OpenStreetMap (ODbL). Routes: OSRM. "~1 in 2" is my own commute log, not company data.

Clickable prototype: SafwatSameer.github.io/smart-dropoff/prototype